

Gettier, Epistemic Closure, and the Necessity of N_1 :

A TNA Critique of Justified True Belief

Abstract

This paper analyzes the Gettier problem through the framework of the Theory of Axiomatic Necessity (TNA). Classical epistemology traditionally defines knowledge as Justified True Belief (JTB). However, the famous counterexamples introduced by Edmund Gettier demonstrate that justification, truth, and belief are insufficient to guarantee genuine knowledge.

This paper argues that the failure of JTB is not accidental but **structural**. Gettier cases reveal a deeper phenomenon identified in TNA as **Failure of Local Closure**: the impossibility of generating complete epistemic legitimacy solely from internally accessible conditions.

Under TNA, genuine knowledge requires not only belief, justification, and truth, but also a **non-derivable realizability connection** between the cognitive structure of the subject and the truth-condition of reality. This missing component is formalized as N_1 : the external legitimizing structure that connects belief to truth.

The Gettier problem therefore becomes a special case of a broader ontological principle: **no sufficiently complex system can generate its own conditions of epistemic legitimacy entirely from internal dynamics**.

1. Introduction

The problem introduced by Edmund Gettier in 1963 remains one of the foundational crises of modern epistemology.

Classically, knowledge was defined as:

$$K = J \wedge T \wedge B$$

where:

- J = justification

- T = truth
- B = belief

This definition appeared stable for centuries because it seemed to capture the essential structure of knowing: one must believe, one must be justified, and the belief must correspond to reality.

However, Gettier demonstrated that all three conditions may hold while genuine knowledge still appears absent.

The implication was devastating: **epistemic coherence alone cannot guarantee legitimate knowledge.**

TNA interprets this not as a minor technical problem, but as evidence of **structural epistemic non-closure.**

2. The Classical Gettier Problem

Consider Gettier's well-known example.

Smith believes: "*The man who will get the job has ten coins in his pocket.*"

Smith is justified because:

- Jones has ten coins, and
- Smith believes Jones will get the job.

However:

- Smith himself unexpectedly gets the job, and
- Smith also happens to have ten coins in his pocket.

Thus:

- The proposition is **true**,
- Smith **believes** it,
- Smith is **justified**,

Yet intuitively: **Smith does not possess genuine knowledge.**

The truth occurred **accidentally**.

3. JTB as Epistemic Closure Attempt

From the TNA perspective, JTB represents an attempt to fully close epistemology within N_o.

3.1 Belief as Internal State

Belief is locally generated within the cognitive system. It operates entirely within the subject's representational architecture — the domain of N_o.

3.2 Justification as Internal Coherence

Justification depends on:

- Evidence,
- Logical consistency,
- Inferential accessibility,
- Rational coherence.

These are all **internally accessible** — they do not require external anchoring beyond the subject's own cognitive resources.

3.3 Truth as Correspondence

Truth is assumed to be **external** — it corresponds to states of affairs in the world, not merely to internal representations.

However, JTB assumes that:

■ *Internal justification plus external truth are sufficient for legitimate knowledge.*

Gettier shows this assumption **fails**.

4. Failure of Local Closure in Epistemology

TNA formalizes the problem structurally.

The issue is not merely that:

- Justification can fail, or
- Reasoning can be flawed.

Rather:

The epistemic system cannot internally guarantee the legitimacy of its own truth-connections.

This is a direct instance of:

Failure of Local Closure

The cognitive system generates:

- Coherent beliefs,
- Valid reasoning,
- Operational justification,

but cannot ensure that the relation between belief and truth is **structurally legitimate**.

Truth may still occur **accidentally**.

5. The Missing Component: N_1

TNA proposes that knowledge requires an additional condition:

$$K \leftrightarrow (J \wedge T \wedge B) \wedge N_1(B)$$

where $N_1(B) \in \{0, 1\}$ represents:

- **Realizability legitimacy** — the belief is structurally connected to truth in a non-accidental way,
- **Non-accidental truth linkage** — the correspondence between belief and reality is grounded in structural necessity, not coincidence,

- **Structurally valid correspondence** — the relation between cognition and world is mediated by a legitimizing architecture,
- **Non-derivable epistemic closure** — the legitimacy of the connection cannot be generated from within the epistemic system alone.

Equivalently, in TNA operator notation:

$$K = \hat{S}_{ep}(\Sigma)$$

where $\hat{S}_{ep}$ is the **epistemic instantiation** of the general TNA selection operator $\hat{S} : M \rightarrow R$, with:

- $M = \Sigma$ (the multiplicity space of admissible beliefs),
- $R = K$ (the space of legitimately realized knowledge),

and $\hat{S}_{ep}$ selects only those $B_i \in \Sigma$ satisfying simultaneously:

- $J(B_i) = 1$ (justified),
- $T(B_i) = 1$ (true),
- $N_1(B_i) = 1$ (realizability legitimate).

Note on notation: $\hat{S}_{ep}$ is not an autonomous operator distinct from $\hat{S}$. It is the **domain-specific realization** of the general TNA operator within epistemic space. All properties of $\hat{S}$ — non-derivability (Axiom O1), non-unitarity (Axiom O2), and discontinuity (Axiom O3) — apply to $\hat{S}_{ep}$.

6. Formal Interpretation

Define Σ as the multiplicity space of admissible beliefs.

The subject constructs $B_i \in \Sigma$ through justification processes.

However, only some beliefs possess **legitimate realizability linkage** to truth.

The epistemic selection operator $\hat{S}_{ep} : \Sigma \rightarrow K$ does not merely select coherent beliefs; it selects **legitimately realizable truth-connections**.

Gettier cases occur when:

- $J(B_i) = 1$ (justification succeeds),

- $T(B_i) = 1$ (truth succeeds),
- $N_1(B_i) = 0$ (realizability legitimacy fails).

That is: justification succeeds, truth succeeds, but **realizability legitimacy fails**.

The subject's cognitive system has generated a belief that happens to be true, but the structural connection between the belief-generation process and the truth-condition is **accidental** — it lacks the N_1 -mediated legitimizing architecture.

7. Epistemic Luck as Structural Signal

The central insight of TNA is that **epistemic luck is not accidental noise**.

It is a **signal**.

Gettier luck reveals: **the absence of legitimate realizability connection**.

Epistemic luck is **phenomenological evidence of missing N_1 structure**.

The "coincidence" present in Gettier cases is precisely the **symptom of non-closure** — the cognitive system has encountered a state where its internal dynamics (belief + justification) have produced a result that aligns with external reality (truth) without the mediating structure (N_1) that would make this alignment **structurally necessary rather than accidentally convergent**.

8. Why Post-Gettier Theories Keep Reintroducing N_1

Most post-Gettier epistemologies implicitly attempt to restore an N_1 -like condition.

THEORY	HIDDEN N_1 MECHANISM
Reliabilism (Goldman, 1979)	External causal legitimacy — truth-tracking via reliable cognitive processes
Tracking theories (Nozick, 1981)	Counterfactual truth linkage — sensitivity and adherence conditions
Virtue epistemology (Sosa, 1991)	Structural cognitive legitimacy — epistemic competence and character
Defeasibility theories	Stability against hidden disruption — immunity to undermining evidence

THEORY	HIDDEN N ₁ MECHANISM
Proper functionalism (Plantinga, 1993)	Correct realizability architecture — cognitive faculties functioning according to design plan

All attempt to solve the same structural problem:

How can truth become non-accidentally connected to cognition?

TNA interprets all these efforts as **incomplete rediscoveries of N₁**. Each theory identifies a different facet of the legitimizing structure — causal, counterfactual, competence-based, stability-based, functional — but none formalizes the underlying principle: **the necessity of an external, non-derivable realizability condition**.

9. Knowledge as Realizability Participation

Under TNA, knowledge is not merely:

- Internal coherence,
- Symbolic manipulation, or
- Inferential correctness.

Knowledge requires participation in realizability structure itself.

Thus:

To know is not only to represent truth, but to be structurally coupled to the conditions under which truth can be non-accidentally realized.

This repositions epistemology closer to:

- Plato (participation in the Forms as structural realizability),
- Aristotle (knowledge as grasping causes, not merely correlations),
- Classical notions of **Logos** (structured rationality grounded in the architecture of reality),

while avoiding simplistic metaphysical commitments.

10. Structural Implications

10.1 Knowledge Cannot Be Fully Self-Generated

No epistemic system can internally certify its own legitimacy. The attempt to close epistemology within N_0 — to generate knowledge solely from belief, justification, and truth as internally accessible conditions — necessarily fails. N_1 must be **externally anchored**.

10.2 AI Systems Face the Same Problem

An AI may generate:

- Coherent outputs,
- Statistically justified predictions,
- Internally valid reasoning,

while still lacking **legitimate realizability linkage**.

This creates **machine analogues of Gettier cases**: an AI may produce a true, justified belief (e.g., a correct prediction) through processes that lack structural connection to the truth-condition. The prediction is correct, the reasoning is valid, the training data supports it — yet the alignment between the AI's representational process and reality remains **operationally coherent but ontologically fragile**.

This connects directly to TNA's analysis of AI in N_0 , N_1 , and *the Structural Origin of Freedom, Evil, Religion, and Artificial Intelligence* (Bresciano, 2025) and *The Selector and the Operator* (Bresciano, 2026): AI possesses the **Operator** (formal mechanisms of selection) but lacks the **Selector** (the phenomenological interface that registers multiplicity as existential tension and triggers the necessity of selection). AI executes without the discharge of necessity.

10.3 Rationality Alone Is Insufficient

Perfect logical structure does not guarantee genuine knowledge. A system may be:

- Internally flawless,
- Externally disconnected.

Rationality is **necessary but not sufficient** for epistemic legitimacy. N_1 provides the bridge between internal coherence and external realizability.

11. Relation to the Block Universe and Consciousness

The Gettier problem connects to broader TNA themes through a shared **structural isomorphism**:

DOMAIN	PHENOMENON	MISSING STRUCTURE
Block Universe	Consciousness experiences unresolved multiplicity across time	N_1 as the bridge between temporal experience and atemporal realizability
Gettier Problem	Cognition experiences unresolved truth-connection	N_1 as the bridge between internal belief and external realizability
Quantum Measurement	Observer experiences unresolved superposition	N_1 as the bridge between quantum multiplicity and classical actuality
Economic Markets	Agents experience unresolved opportunity multiplicity	N_1 as the bridge between preference and realized exchange

In the Block Universe (Bresciano, 2025), consciousness confronts the atemporal totality of events as a multiplicity that cannot be internally resolved — the subject cannot generate the conditions of temporal passage from within the block structure. The N_1 bridge is required: a **non-derivable, non-unitary, discontinuous** structure that connects the local experience of time to the global realizability of events.

In the Gettier problem, cognition confronts a multiplicity of possible truth-connections (the belief could be true for the reasons Smith believes, or for accidental reasons) that cannot be internally resolved — the subject cannot generate the conditions of non-accidental truth from within the JTB structure. The N_1 bridge is required: the same **non-derivable, non-unitary, discontinuous** structure that connects local belief to global realizability.

The isomorphism is exact:

- Block Universe: temporal experience :: Gettier: epistemic experience
- Unresolved temporal multiplicity :: Unresolved epistemic multiplicity
- N_1 as temporal realizability bridge :: N_1 as epistemic realizability bridge
- Failure of local temporal closure :: Failure of local epistemic closure

This is not analogy. It is **structural identity across domains** — the signature of TNA as a transdisciplinary framework.

The connection to consciousness is direct: in *Consciousness as the N_1 – N_0 Bridge* (Bresciano, 2025), consciousness is identified as the **stability** of the $N_1 \rightarrow N_0$ bridge — the phenomenological continuity that maintains the self across discrete selection events. In epistemology, this same stability manifests as the **confidence** that one's knowledge is not merely accidentally true — the

"feeling of knowing" that accompanies (but does not constitute) genuine knowledge. When N_1 is present, the Selector (consciousness) registers the discharge of necessity; when N_1 is absent, the Selector registers only the tension of unresolved multiplicity — the "hunch" that something is epistemically amiss, even when justification and truth appear satisfied.

12. Final Implication

The Gettier problem is not merely a technical objection to JTB.

It is evidence of a deeper ontological principle:

■ *Legitimate realizability cannot be completely generated from internal structure.*

Knowledge therefore requires:

- Belief,
- Justification,
- Truth,
- **And participation in a legitimizing structure beyond local epistemic dynamics.**

The failure of JTB is thus not accidental.

It is the **epistemological manifestation of Failure of Local Closure**.

Conclusion

TNA interprets the Gettier problem as a **structural failure of epistemic closure**.

Classical epistemology attempted to define knowledge entirely through internally accessible conditions: belief, justification, and truth.

Gettier demonstrated that this closure **fails**.

Truth may occur accidentally even within perfectly coherent epistemic systems.

TNA explains why: because genuine knowledge requires an additional **non-derivable realizability condition**:

$$K \leftrightarrow (J \wedge T \wedge B) \wedge N_1(B)$$

Without this condition:

- Justification becomes operationally coherent but ontologically fragile,
- Truth becomes accidental,
- Knowledge collapses into epistemic luck.

The Gettier problem therefore reveals something profound:

■ *Cognition alone cannot fully guarantee its own legitimacy.*

Knowledge requires participation in a larger structure of realizability that exceeds local epistemic dynamics — the N_1 structure that connects the finite subject to the architecture of reality itself.

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